

Package #155

replace conventional aluminum oxide grinding wheel with green silicon carbide wheel for titanium alloy grinding

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented manufacturing problem: when grinding titanium alloys (Ti-6Al-4V) with conventional aluminum-oxide wheels, titanium chips adhere to the wheel's abrasive grains and bond (wheel loading); the loaded wheel stops cutting, friction rises, the wheel glazes, and the workpiece surface temperature spikes — producing tensile residual stresses, alpha-case burn, microcracking, and scrapped expensive aerospace parts. Titanium's low thermal conductivity and chemical reactivity with abrasive make this worse than steel grinding. Qualitative engineering-literature context; no fabricated statistics (retrieval must custody the evidence).

Mechanism: green silicon carbide wheel maintained better thermal stability and wear resistance

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.